popquiz Newton's applications and bonus / test2

This is a preview of the published version of the quiz

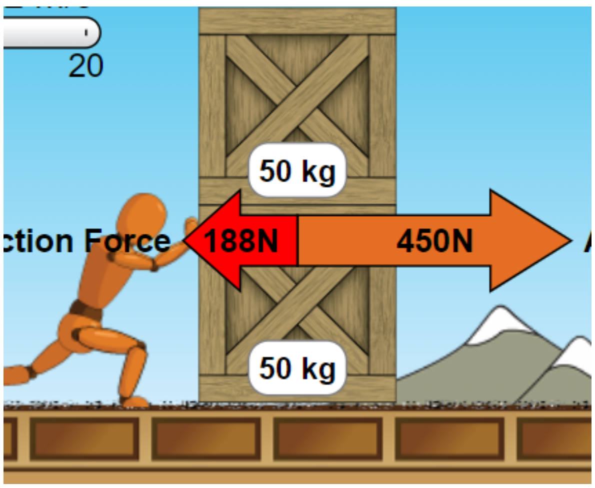
Started: Apr 29 at 7:36pm

Quiz Instructions

bonus points !!

review for test

Question 1 20 pts



Here is a 2 objects system. 2 boxes on top of each other. The dummy is applying a force on the bottom box. the 2 boxes move together. What is the acceleration of the system . There is kinetic friction as shown. hint: in the ": bag" you have the 2 boxes. The forces (applied force and friction) are acting on the system 2 boxes. (100kg)

- 1.88m/s/s
- 4.5 m/s/s
- 5.24m/s/s
- 2.62m/s/s

Question 2 20 pts

Building on the previous question. Consider the top crate only (50kg). So the system (in a "bag") is the top crate only.

what is true?

hint: if a system accelerates. that means there is a net force in the same direction as the acceleration. dah

and the acceleration is @ right. Also. Here the net force. Is not the dummy push. The dummy does not push the top box. It does not touch it.

also. static means : it does not slide

Its wood against wood.

https://phet.colorado.edu/sims/html/forces-and-motion-basics/latest/forces-and-motion-basics_en.html

- The top mox is only acted upon by its weight and normal froce (support). no friction
- there is no force acting on the top box
- the top box is acted by kinetic friction @ right
- the top box is acted by static friction @ left
- The top box is acted upon by a static frictional force @ right

Question 3 20 pts

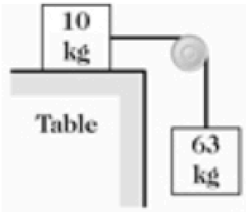
Building on the previous question. What is the force acting on the top box?

hint: place the top box (50kg) in an imagninary box. Its acceleration is the same as in Q1. The force applied to is the static friction.

- 450N
- 131N
- 0N
- 262N

Question 4 20 pts

As shown in the figure, a 10-kg block on a perfectly smooth horizontal table is connected by a horizontal string to a 63-kg block that is hanging over the edge of the table. What is the magnitude of the acceleration of the 10-kg block when the other block is gently released?



- [ ] it can't move
- [ ] 9m/s/s
- [ ] 7.5m/s/s
- [ ] 8.6 m/s/s

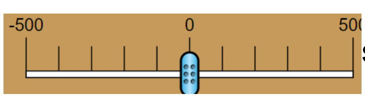
Question 5 20 pts

go to:

[https://phet.colorado.edu/sims/html/forces-and-motion-basics/latest/forces-and-motion-basics_en.ht (https://phet.colorado.edu/sims/html/forces-and-motion-basics/latest/forces-and-motion-basics_en.ht)](https://phet.colorado.edu/sims/html/forces-and-motion-basics/latest/forces-and-motion-basics_en.ht)

click on acceleration. check everything.

PLace two crates on top of each other as shown in Q1. PLace the mouse of the dummy and have it push unti it starts moving. slooooooowly. you can also use the slider:



sloooooooowly

What is the maximum static friction? (before kinetic friction takes over)

- [ ] about 75N
- [ ] about 250N
- [ ] about 150N

about 125N

Question 6 5 pts

bonus

building on the previous question.

static friction max = coefficient x normal. Whats the coefficient of static friction

0.5

o

0.25

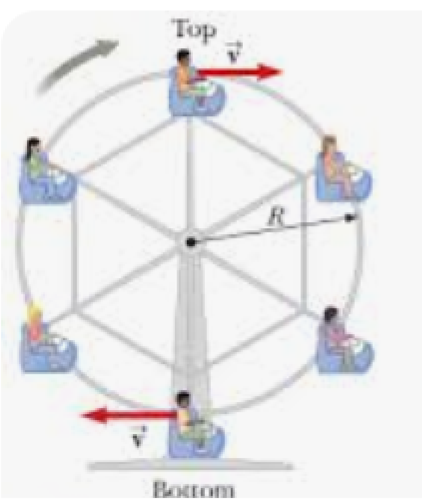
0,1

Question 7 5 pts

another bonus!!

Oscar is at the top of a Ferris wheel thats moving at a speed of 5m/s

the radius of the wheel is 20m. The mass of Oscar is 100kg. g=10m/s/s



whats his apparent weight (find the normal. dah).

acceleration is down (toward center) = V^2 / R

so positive direction is down.

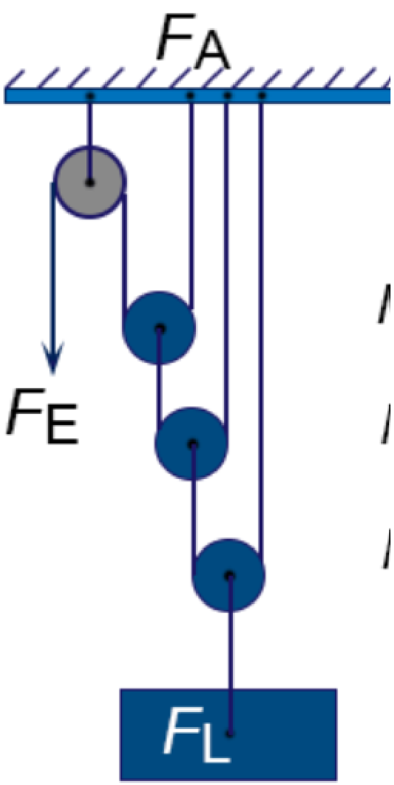
down - up = m V^2 / R

down is the weight. up is the Normal (from sit to kid). Find the normal.

- 1000N
- 0N
- 875N
- 1125N

Question 8 5 pts

another bonus question. good review for the test!



Whats the mechanical advantage ?

hint: suppose the load is 400 lbs. proceed from bottom pulley (in a bag) then the one above then the one above. MA = load/effort

- 2:1

- 4:1
- 1:1
- 8:1

Not saved Submit Quiz